

SDP 302

Problem Set: Efficient Market Hypothesis

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Instructions & Information

According to the **Efficient Market Hypothesis**, asset prices in an efficient financial market reflect all currently available information. It follows that prices change in response to new information. Because genuinely new information is unpredictable, these price changes should also be unpredictable and independent of past returns. If past returns could be used to predict future returns, traders would exploit this pattern until it disappeared.

These ideas can be expressed through a mathematical model in which log prices follow a **random walk**:

$$\ln(P_t) = \ln(P_{t-1}) + \epsilon_t, \quad \epsilon_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2).$$

Here, P_t is the price at time t , and ϵ_t is a random change. The notation “i.i.d.” means that these changes are independent and follow the same probability distribution. In this model, that distribution is normal, with a mean of zero and variance σ^2 .

Using log prices rather than prices themselves ensures that the model cannot generate negative asset prices. A random walk applied directly to prices could eventually produce a negative value, which would not be meaningful for standard assets such as shares or bonds. Logarithmic notation has a further advantage: log returns are additive over time and, according to this model, should follow a normal distribution:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) = \ln(P_t) - \ln(P_{t-1}) = \epsilon_t$$

Despite being key to the theoretical foundations of **quantitative finance**, the random walk model has significant limitations. In particular, it fails to reproduce many of the patterns and statistical properties observed in real-world prices and returns across different types of assets.

Empirical evidence shows that asset returns actually depart from a normal distribution. The **Jarque–Bera test** can be used to assess whether sample data are normally distributed. It does so by considering the sample size n and two features of the distribution: skewness S , which measures asymmetry, and kurtosis K , which measures the weight of the tails. The test statistic is calculated as follows:

$$JB = \frac{n}{6} \left(S^2 + \frac{(K - 3)^2}{4} \right)$$

Under the null hypothesis of normality, the JB statistic approaches a χ^2 distribution with two degrees of freedom as the sample size increases. A sufficiently large value of JB therefore provides evidence against the null hypothesis. The relevant critical values are 5.99 at the 5% significance level and 9.21 at the 1% significance level. Other widely used tests for normality include the **Shapiro–Wilk test**, the **Kolmogorov–Smirnov test** and the **Anderson–Darling test**.

[EXR] Exercise 1 (Empirical Stock Returns & Tests for Normality)

The file `treet.parquet` on *Canvas* contains daily closing share prices for *Treet Corporation* on the Karachi Stock Exchange from 1 January 2010 to 2 September 2026. The data were obtained from *Investing.com*. Import the file into a `pandas.DataFrame` using the `pandas.read_parquet()` method. Then explore the data and evaluate the *Efficient Market Hypothesis* by answering the questions below.

The *Jarque-Bera test* is available in the `statsmodel` library. After importing the library, you can run the test using the method `jarque_bera()`.

- a) What type of data are these: cross-sectional, panel or time-series data?
- b) Most variables in the sample contain $n = 4110$ observations, whereas the Log Returns column contains only $n = 4109$. Explain this difference.
- c) Use a line plot to visualise how log prices change over time. Create a second line plot showing how log returns change over time.
- d) Plot a histogram to visualise the distribution of log returns. If possible, overlay a curve showing the corresponding normal distribution.
- e) Calculate both the skewness and the kurtosis of log returns. Based on these values would you say log returns are normally distributed?
- f) Perform the Jarque-Bera test and report the *JB* test statistic. Does the result provide sufficient evidence to reject the null hypothesis that log returns follow a normal distribution? Explain your conclusion with reference to the appropriate significance level.